

Challenge (Habanero)

- Q1.** A recipe book contains 24 recipes, with 12 being vegetarian and 8 having a cooking time of over 30 minutes. If 4 recipes are both vegetarian and take over 30 minutes to cook, what is the probability that a randomly selected recipe is either vegetarian or takes over 30 minutes to cook?
- (A) $\frac{5}{6}$
(B) $\frac{2}{3}$
(C) $\frac{3}{4}$
(D) $\frac{1}{2}$
(E) $\frac{1}{3}$
- Q2.** A construction company builds houses with either a flat or a sloping roof. The relative frequencies of houses with flat and sloping roofs in a particular neighborhood are 0.4 and 0.6, respectively. If 30 houses have flat roofs, how many houses are there in the neighborhood?
- (A) 50
(B) 60
(C) 70
(D) 75
(E) 80
- Q3.** A restaurant owner wants to analyze the association between the type of cuisine and the day of the week. The two-way table below shows the frequency of customers who order Italian, Mexican, or Indian food on Monday, Tuesday, or Wednesday. What is the relative frequency of customers who order Italian food on Monday?
- (A) $\frac{1}{10}$
(B) $\frac{1}{5}$
(C) $\frac{1}{3}$
(D) $\frac{2}{5}$
(E) $\frac{3}{5}$
- Q4.** A baker sells 240 loaves of bread per day, with 120 being whole wheat and 80 being white bread. If 40 customers buy both whole wheat and white bread, what is the probability that a randomly selected customer buys either whole wheat or white bread?
- (A) $\frac{3}{4}$
(B) $\frac{2}{3}$
(C) $\frac{3}{5}$
- (D) $\frac{1}{2}$
(E) $\frac{1}{3}$
- Q5.** A chef wants to analyze the relationship between the method of cooking and the type of meat used. The two-way table below shows the frequency of chicken, beef, and pork cooked using grilling, roasting, or sautéing. What is the relative frequency of beef cooked using roasting?
- (A) $\frac{1}{6}$
(B) $\frac{1}{4}$
(C) $\frac{1}{3}$
(D) $\frac{2}{5}$
(E) $\frac{1}{2}$
- Q6.** A contractor builds houses with either 2, 3, or 4 bedrooms. The relative frequencies of houses with 2, 3, and 4 bedrooms are 0.2, 0.5, and 0.3, respectively. If 120 houses have 3 bedrooms, how many houses are there in total?
- (A) 150
(B) 200
(C) 240
(D) 250
(E) 300
- Q7.** A recipe calls for 2 cups of flour, 1 cup of sugar, and 0.5 cups of milk. If 1 cup of flour weighs 120 grams, 1 cup of sugar weighs 200 grams, and 1 cup of milk weighs 240 grams, what is the total weight of the ingredients?
- (A) 480 grams
(B) 520 grams
(C) 600 grams
(D) 720 grams
(E) 820 grams
- Q8.** A builder uses either 10mm, 15mm, or 20mm pipes for plumbing. The relative frequencies of 10mm, 15mm, and 20mm pipes are 0.3, 0.4, and 0.3, respectively. If 60 pipes are 15mm, how many pipes are there in total?
- (A) 100
(B) 120
(C) 150
(D) 180
(E) 200

Open Ended Questions (FRQ)

- Q9.** A two-way table shows the frequency of houses with 1, 2, or 3 stories and the type of roofing material used (asphalt, metal, or tile). The table is as follows:
- | | Stories | Asphalt | Metal | Tile | Total |
|-------|---------|---------|-------|------|-------|
| 1 | 20 | 10 | 15 | 45 | |
| 2 | 30 | 20 | 10 | 60 | |
| 3 | 15 | 25 | 20 | 60 | |
| Total | 65 | 55 | 45 | 165 | |
- Calculate the relative frequency of houses with 2 stories and metal roofing.
- Q10.** A chef wants to analyze the association between the type of dessert (cake, pie, or ice cream) and the day of the week (Monday, Tuesday, or Wednesday). The two-way table below shows the frequency of customers who order each type of dessert on each day.
- | | Dessert | Monday | Tuesday | Wednesday | Total |
|-----------|---------|--------|---------|-----------|-------|
| Cake | 15 | 20 | 10 | 45 | |
| Pie | 10 | 15 | 20 | 45 | |
| Ice Cream | 20 | 10 | 15 | 45 | |
| Total | 45 | 45 | 45 | 135 | |
- Calculate the probability that a customer orders cake on Tuesday.

Chef's Tips

- Chef's Tips: Read each question carefully and choose the correct answer.
- Workings and explanations are not required for multiple-choice questions.
- For free-response questions, show all workings and explanations.